Identifying Tracking Motion Pattern in Rehabilitation Using Inverse Optimal Control

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***Abstract*—Rehabilitation training is essential for hemiplegia recovery and "tracking" is one of the most fundamental rehabilitation tasks. Nevertheless, the tracking motions between the patients are highly personalized. To maximize the treatment potency for each patient, it requires us to understand and model the patient's personalized tracking pattern during the rehabilitation training task. In this paper, we model the behaviour of a patient tracking a given reference signal by a linear quadratic optimal problem, while her control inputs are corrupted by signal-dependent noise. Since the parameters in the objective function stand for the trade-off between tracking error and control effort, they are interpreted as the "personalized" tracking motion pattern and the goal is to identify them based on the observed data. Such a problem is called Inverse Optimal Control (IOC). We first prove that the model structure is identifiable. Then we propose a statistically consistent IOC algorithm that handles multiplicative noise induced by muscle fluctuations. The performance of the objective function estimation is then illustrated by both numerical simulations and experiments.**

***Index Terms*—inverse optimal control, signal-dependent noise,rehabilitation robot**

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I. Introduction

Rehabilitation training is essential for treating hemiplegia caused by stroke and "tracking" is one of the most fundamental rehabilitation tasks [1]. It also helps to reconstruct the hand-eye coordination. In addition, mirror therapy [2] is a widely used clinical treatment. In particular, the patient is asked to hide her affected side limb behind a mirror and let the non-affected side move. The patient sees the reflection of the non-affected side limb in the mirror and imagines it as if the affected side limb is moving accordingly. Inspired by this, we suspect that a similar tracking motion pattern to her own non-affected side would give a better treatment potency. Thus we are motivated to model the patient's personalized tracking pattern and hence use the model for rehabilitation robot controller design in the future. Ideally, once the personalized tracking motion model is built, we hopefully can know how much assistance the rehabilitation robot needs to offer towards the patient during rehabilitation training.

To this end, we focus on the following scenario in rehabilitation training: the patient rotates a crank (see Fig. 1) whose angular position and velocity are mapped to pointers shown on the screen. Given a reference signal, the patient is then asked to track the reference signal by rotating the crank. Since the dynamics of the crank is linear, we model the human's tracking motion as a linear quadratic optimal tracking control. Note that, however, larger intended driving force would lead to larger muscle fluctuations [3]. This instinct implies that the dynamics in the linear quadratic optimal tracking control would involve signal-dependent process noise. Despite the challenge, most importantly, the parameters in the objective function stand for the trade-off between the tracking error and control effort, they are interpreted as the "personalized" tracking motion pattern. This further motivates us to do Inverse Optimal Control (IOC) so as to identify the patient's underlying "personalized" parameters in the objective function using observed data.

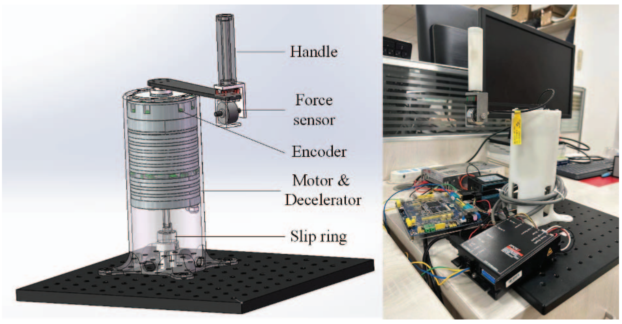

Fig. 1. The crank-shaped rehabilitation robot design. The force sensor can measure the one-dimensional force that is tangential to the rotation. This is done by a special mechanical design.

IOC problems have received considerable attention in recent years. In particular, linear quadratic cases are studied in both continuous-time [4]–[7] and discrete-time [8]–[10] scenarios. However, tracking problems are not considered in the aforementioned papers. Consequently, linear quadratic tracking IOC problems with a stochastic time-horizon length are considered in the second author's recent work [11], [12], yet the dynamics does not involve signal-dependent process noise and hence can not be applied directly here. On the other hand, more general nonlinear IOC problems (non-quadratic objective function and nonlinear dynamics) have been considered in [13]–[18]. However, the effect of noise is not well-considered in this more general work. It is even proved in [19] that the methods proposed in [13], [17], [18] are not statistically consistent. Meanwhile, since the patient can choose when to start tracking once the reference signal is released, the IOC algorithm needs to be able to handle such time-horizon length stochasticity like [11], [12]. Therefore, we are motivated to extend the second author's previous work [11], [12] and obtain an accurate and robust model estimate.

In this work, we formulate the tracking motion pattern identification problem in rehabilitation training as a linear quadratic IOC problem. In particular, the linear dynamics involves signal-dependent process noise. We first show the identifiability of such model structure and develop a statistically consistent IOC algorithm that can handle the signal-dependent process noise. The performance of the objective function estimate is then illustrated by both numerical simulations and experiments.

Notations: We use $\mathbb{S}_+^n$ to denote the set of positive semi-definite matrices with dimension n. For integers a_1, a_2, we denote $a_1 : a_2$ as $a_1, a_1+1, \dots, a_2$. $\|\cdot\|_F$ and $\|\cdot\|$ denote Frobenius norm and l_2 norm, respectively. We use ***italic bold font*** to highlight stochastic elements.

II. Problem Formulation

In a rehabilitation training task for stroke patients with hemiplegia, the patient is asked to sit in front of a computer screen. In addition, there is a pointer shown on the screen which is controlled by the rotation of the crank (see Fig. 1). The patient is then asked to rotate the crank so that the pointer tracks a given reference signal shown on the screen. Since patients differ a lot in their arm length, body weight, etc., it is hard to give a precise musculoskeletal dynamical model (e.g., see [20]) for each patient in practice. Therefore, we focus on describing the motion of the crank-shaped rehabilitation robot under the patient's manipulation instead.

To this end, let $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathbb{P}})$ be a probability space that carries random vectors $\boldsymbol{x}, \boldsymbol{u} \in \mathbb{R}^2$, $\boldsymbol{w} \in \mathbb{R}^2$. The dynamics of the crank-shaped rehabilitation robot is given by

$$d\boldsymbol{x} = \begin{bmatrix} 0 & 1 \\ \frac{-k}{m} & \frac{-b}{m} \end{bmatrix} \boldsymbol{x}dt + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} (\boldsymbol{u}^1 dt + \boldsymbol{u}^2 dt + \hat{\sigma}\boldsymbol{u}^1 d\boldsymbol{w}), \quad (1)$$

where m, k and b are the moment of inertia, elasticity and viscosity coefficients yield by the mechanical structure and motor. In addition, $\boldsymbol{x}(t) = [\boldsymbol{x}_1(t), \boldsymbol{x}_2(t)]^T \in \mathbb{R}^2$ denotes the angular position and velocity of the crank joint, respectively. $\boldsymbol{w}$ denotes a standard one-dimensional Brownian motion and $\boldsymbol{u}^1$ is the patient's intended driving torque. Since larger intended driving torque would lead to larger fluctuations [3], such phenomenon caused by muscle actuation is modelled as the Brownian motion scaled by $\hat{\sigma}\boldsymbol{u}^1$. Moreover, $\boldsymbol{u}^2$ is the torque provided by the motor.

Since the motor in Fig. 1 is equipped with a high-ratio gear box, it is physically not possible for the user to directly rotate it without any controller design on $\boldsymbol{u}^2$ when the motor is on. To this end, we first apply an admittance controller to $\boldsymbol{u}^2$. This also add some "softness" to the crank and provides a safety guarantee and not hurt the user. In particular, let

$$\bar{A} = \begin{bmatrix} 0 & 1 \\ \bar{a}_1 & \bar{a}_2 \end{bmatrix}, \quad \bar{B} = \begin{bmatrix} 0 \\ \bar{b}_1 \end{bmatrix}$$

be the desired system dynamics (with a proper elasticity, viscosity and moment of inertia). Since the actual input by human $\boldsymbol{u}^1 + \hat{\sigma}\boldsymbol{u}^1\boldsymbol{w}$ can be measured by the force sensor, the admittance controller $\boldsymbol{u}^2$ takes the form

$$\boldsymbol{u}^2 = (m\bar{a}_1 + k)\boldsymbol{x}_1 + (m\bar{a}_2 + b)\boldsymbol{x}_2 + (m\bar{b}_1 - 1)(\boldsymbol{u}^1 + \hat{\sigma}\boldsymbol{u}^1\boldsymbol{w}). \quad (2)$$

Hence the crank dynamics during the data collection process becomes $d\boldsymbol{x} = \bar{A}\boldsymbol{x}dt + \bar{B}(\boldsymbol{u}^1 dt + \boldsymbol{u}^1 d\boldsymbol{w})$. Next, we discretize the dynamics since conducting the discrete algorithm in a computational system is more natural. To this end, let Δt be the sampling period, and assuming $\boldsymbol{u}^1$ and $\boldsymbol{u}^2$ stays the same in the short sampling period, we discretize the system dynamics (1) as

$$\begin{aligned} \boldsymbol{x}(t_{k+1}) = \underbrace{e^{\bar{A}\Delta t}}_{A} \boldsymbol{x}(t_k) + \underbrace{\int_0^{\Delta t} e^{\bar{A}s}\bar{B}ds}_{B}\, \boldsymbol{u}^1(t_k) \\ + \underbrace{e^{\bar{A}\Delta t}\bar{B}}_{G} \boldsymbol{u}^1(t_k)\boldsymbol{w}(t_k), \quad (3) \end{aligned}$$

where we approximate the Itô intergral by first order and $\boldsymbol{w}(t_k) \sim \mathcal{N}(0, \sigma^2 := \hat{\sigma}^2\Delta t)$ [21]. To abbreviate the notations, we denote $\boldsymbol{x}_k := \boldsymbol{x}(t_k)$, $\boldsymbol{u}_k := \boldsymbol{u}^1(t_k)$ and $\boldsymbol{w}_k := \boldsymbol{w}(t_k)$ from now on. In addition, by the evidence that human trade-off between the error and torque cost during tracking [11], given a reference signal r_k, $k \in [1, \nu]$, we formulate the optimal criteria of patient's tracking motion as

$$\begin{aligned} J = \mathbb{E}\Big\{ \sum_{k=\nu-N+1}^{\nu-1} \Big[\frac{1}{2}(\boldsymbol{x}_\nu - r_\nu)^T \bar{Q}(\boldsymbol{x}_\nu - r_\nu) \\ \frac{1}{2}(\boldsymbol{x}_k - r_k)^T \bar{Q}(\boldsymbol{x}_k - r_k) + \frac{1}{2}\|\boldsymbol{u}_k\|^2\Big]\Big\}, \quad (4) \end{aligned}$$

where $\bar{Q} = \mathrm{diag}(\bar{q}_1, \bar{q}_2)$. The unknown matrix $\bar{Q} \in \mathbb{S}_+^n$ describes the trade-off between tracking error and torque cost in the rehabilitation tracking task. In addition, given a reference signal $\{r_k\}_{k=1}^{\nu}$, the patient can choose when to start tracking with random initial position for each trial. To this end, let $(\Omega, \mathcal{F}, \mathbb{P})$ be the probability space that carries the discretized random vector sequence $\{\boldsymbol{w}_k\}_{k=1}^{\nu-1}$, random vector $\bar{\boldsymbol{x}}$ and a random variable $\boldsymbol{N} = \{2, 3, \dots, \nu\}$. Thus, given a realization $(\bar{x}, N)$ of $(\bar{\boldsymbol{x}}, \boldsymbol{N})$, the states of the crank under the patient's manipulation is governed by

$$\begin{aligned} \min_{\substack{\boldsymbol{x}_{1:\nu} \\ \boldsymbol{u}_{1:\nu}}} \mathbb{E}_{\boldsymbol{w}_{\nu-N+1:\nu}} \Big[\frac{1}{2}\begin{bmatrix}\boldsymbol{x}_\nu^T & r_\nu^T\end{bmatrix} \bar{\mathcal{Q}}_\nu(\bar{Q}) \begin{bmatrix}\boldsymbol{x}_\nu \\ r_\nu\end{bmatrix} + \\ \sum_{k=\nu-N+1}^{\nu-1} \frac{1}{2}\begin{bmatrix}\boldsymbol{u}_k^T & \boldsymbol{x}_k^T & r_k^T\end{bmatrix} \bar{\mathcal{Q}}(\bar{Q}) \begin{bmatrix}\boldsymbol{u}_k \\ \boldsymbol{x}_k \\ r_k\end{bmatrix}\Big] \quad (5a) \end{aligned}$$

$$\text{s.t.} \quad (3) \quad (5b)$$

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k, \boldsymbol{u}_k = 0, \quad k = 1 : \nu - N, \quad (5c)$$

$$\boldsymbol{x}_1 = \bar{x}, \quad (5d)$$

where $\bar{\mathcal{Q}}_\nu(\bar{Q}) := \begin{bmatrix} \bar{Q} & -\bar{Q} \\ -\bar{Q} & \bar{Q} \end{bmatrix}$, $\bar{\mathcal{Q}}(\bar{Q}) := \begin{bmatrix} I & 0 & 0 \\ 0 & \bar{Q} & -\bar{Q} \\ 0 & \bar{Q}^T & \bar{Q} \end{bmatrix}$ and (5c) describe the trajectories as well as the control before the patient starts tracking. We refer (5) as "forward" problem in the remainder of this paper.

Before we proceed, we make the following mild assumptions regarding the stochastic elements as well as the parameter $\bar{Q}$ to be estimated. The assumptions are almost the same as the ones in [11].

Assumption II.1 (I.I.D random variables) *$\{\boldsymbol{w}_k\}_{k=1}^{\nu-1}$ and $(\bar{\boldsymbol{x}}, \boldsymbol{N})$ are mutually independent. In addition, $\{\boldsymbol{w}_k\}_{k=1}^{\nu-1}$ are zero-mean white Gaussian noise, where $\boldsymbol{w}_k \sim \mathcal{N}(0, \sigma^2), \forall k = 1 : \nu - 1$.*

Assumption II.2 (Support of the planning horizon) *It holds that $\nu \geq 2$, $\mathbb{P}(\boldsymbol{N} \in [2, \nu]) = 1$ and $\mathbb{P}(\boldsymbol{N} = \nu) > 0$.*

Assumption II.3 (Persistent excitation) *It holds that $\mathrm{cov}_{\boldsymbol{x}_k|\boldsymbol{N}=\nu}(\boldsymbol{x}_k, \boldsymbol{x}_k) \succ 0$ and $\mathbb{E}[\|\boldsymbol{x}_k\|^2] < \infty$, for all $k = 1 : \nu$.*

Assumption II.4 (Bounded parameter) *The "true" $\bar{Q}$ that remains to be estimated in (5a) lives in a compact set*

$$\mathbb{G}(\varphi) = \{\bar{Q} \in \mathbb{S}_+^n | \|\bar{Q}\| \leq \varphi\}.$$

Let $\boldsymbol{x}_k^i$ and $\boldsymbol{\mu}_k^i$ be identically distributed as $\boldsymbol{x}_k$ and $\boldsymbol{\mu}_k := \boldsymbol{u}_k + \boldsymbol{u}_k \boldsymbol{w}_k$, respectively, for all $i = 1 : M$. Suppose that we observe M trajectories and the actual torque input by the patient, i.e., $\{x_k^i\}_{k=1}^{\nu}$ and $\{\mu_k^i\}$ $i = 1 : M$, where x_k^i and μ_k^i are just realizations of $\boldsymbol{x}_k^i$ and $\boldsymbol{\mu}_k^i$. We aim to identify the parameter $\bar{Q} = \mathrm{diag}(\bar{q}_1, \bar{q}_2)$ based on the observations. In addition, note that the potentially unknown φ in Assumption II.4 shall not cause any problem since we can always take φ arbitrarily large in practice.

III. Main Results

In this section, we construct the IOC algorithm to identify the patient's motion pattern, namely $\bar{Q}$. In particular, the algorithm is similar to [11], [12], except that we need to deal with the signal-dependent process noise. First, note that given a realization $(\bar{x}, N)$, the optimal control of the "forward" problem (5) can be derived similarly as [22, Sec. 4.1]. And the optimal control $\boldsymbol{u}_k$ ($k = \nu - N + 1 : \nu - 1$) is given by

$$\boldsymbol{u}_k = \underbrace{-\bar{\mathfrak{R}}_k^{-1} \bar{\mathfrak{S}}_k}_{K_k(\bar{Q})} \boldsymbol{x}_k - \bar{\mathfrak{R}}_k^{-1} \bar{g}_k = \underbrace{-\bar{\mathfrak{R}}_k^{-1} \begin{bmatrix} \bar{\mathfrak{S}}_k & \bar{g}_k \end{bmatrix}}_{\tilde{K}_k(\bar{Q})} \begin{bmatrix} \boldsymbol{x}_k \\ 1 \end{bmatrix}, \quad (6)$$

where $\bar{\mathfrak{R}}_k := B^T \bar{P}_{k+1} B + \sigma^2 G^T \bar{P}_{k+1} G + I$, $\bar{\mathfrak{S}}_k := B^T \bar{P}_{k+1} A$, $\bar{g}_k := B^T \bar{\eta}_{k+1}$, and the sequences of $\bar{P}_{1:\nu}$ and $\bar{\eta}_{1:\nu}$ are generated by the Riccati iterations:

$$\bar{P}_\nu = \bar{Q}, \bar{P}_k = \bar{Q} + A^T \bar{P}_{k+1} A - \bar{\mathfrak{S}}_k^T \bar{\mathfrak{R}}_k^{-1} \bar{\mathfrak{S}}_k; \quad (7a)$$

$$\bar{\eta}_\nu = -\bar{Q} r_\nu, \bar{\eta}_k = -\bar{Q} r_k + \left(A - B \bar{\mathfrak{R}}_t^{-1} \bar{\mathfrak{S}}_k\right)^T \bar{\eta}_{k+1}. \quad (7b)$$

Notably, if we see $\begin{bmatrix} \boldsymbol{x}_k^T & 1 \end{bmatrix}^T$ as the input to the optimal controller and $\boldsymbol{\mu}_k := \boldsymbol{u}_k + \boldsymbol{u}_k \boldsymbol{w}_k$ as the output, then the control gain $\tilde{K}_k(\bar{Q})$ in (6) can be seen as the model structure that defines the input-output relationship. Based on this, we give the following lemma regarding the identifiability.

Lemma III.1 *Under Assumption II.2, the model structure $\tilde{K}_k(\bar{Q})$ is strictly globally identifiable.*

Proof *Assume that there exists $Q' \neq \bar{Q}$ that lead to the same model structure $\{\tilde{K}_k(\bar{Q})\}_{k=1}^{\nu-1}$. In view of* (6)*, it follows that*

$$K_{\nu-1}(\bar{Q}) = -\mathfrak{R}_{\nu-1}^{-1} \mathfrak{S}_{\nu-1} = K_{\nu-1}(Q') = -\mathfrak{R}_{\nu-1}'^{-1} \mathfrak{S}_{\nu-1}'.$$

In view of the fact that $P_\nu = \bar{Q}$, $P_\nu' = Q'$ and the expressions of $\bar{\mathfrak{R}}_{\nu-1}$, $\mathfrak{R}_{\nu-1}'$, $\bar{\mathfrak{S}}_{\nu-1}$, $\mathfrak{S}_{\nu-1}'$, the above equation implies that $\mathrm{rank}(\mathcal{K}_{\nu-1}) = 1$ where

$$\mathcal{K}_{\nu-1} = \begin{bmatrix} \bar{\mathfrak{R}}_{\nu-1}^T & \mathfrak{R}_{\nu-1}'^T \\ \bar{\mathfrak{S}}_{\nu-1}^T & \mathfrak{S}_{\nu-1}'^T \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & b_1^2 + g_1^2 \sigma^2 & b_2^2 + g_2^2 \sigma^2 \\ 0 & a_1 b_1 & a_3 b_2 \\ 0 & a_2 b_1 & a_4 b_2 \end{bmatrix}}_{M_{id}} \underbrace{\begin{bmatrix} 1 & 1 \\ \bar{q}_1 & q_1' \\ \bar{q}_2 & q_2' \end{bmatrix}}_{\Gamma(\bar{Q}, Q')}.$$

On the other hand, it is clear that

$$\det\left(\begin{bmatrix} a_1 b_1 & a_3 b_2 \\ a_2 b_1 & a_4 b_2 \end{bmatrix}\right) = \underbrace{(a_1 a_4 - a_2 a_3)}_{\det(A)} b_1 b_2 \neq 0$$

since $\det(A) \neq 0$ and $b_1, b_2 \neq 0$ are guaranteed by the discretization (3)*. This implies that M_{id} is invertible. Moreover, since $\bar{Q} \neq Q'$, $\Gamma(\bar{Q}, Q')$ is of full-column rank, this implies that $\mathrm{rank}(\mathcal{K}_{\nu-1}) = 2$, which is a contradiction.*

Now that we know the model structure is identifiable, we are ready to design the IOC algorithm that can handle the signal-dependent noise. To this end, we know that the Riccati iterations (7) are actually derived from Bellman iterations with value functions $V_k(\boldsymbol{x}_k) = \frac{1}{2} \boldsymbol{x}_k^T \bar{P} \boldsymbol{x}_k + \bar{\eta}_k^T \boldsymbol{x}_k + \bar{\gamma}_k$, where

$$V_k(\boldsymbol{x}_k) = \frac{1}{2} \boldsymbol{x}_k^T \bar{P}_k \boldsymbol{x}_k + \bar{\eta}_k^T \boldsymbol{x}_k + \bar{\gamma}_k = \frac{1}{2} \begin{bmatrix} \boldsymbol{u}_k^T & \boldsymbol{x}_k^T & r_k^T \end{bmatrix} \bar{\mathcal{Q}} \begin{bmatrix} \boldsymbol{u}_k \\ \boldsymbol{x}_k \\ r_k \end{bmatrix}$$
$$+ \mathop{\mathbb{E}}_{\boldsymbol{w}_k} [\frac{1}{2} \boldsymbol{x}_{k+1}^T \bar{P}_{k+1} \boldsymbol{x}_{k+1} + \bar{\eta}_{k+1}^T \boldsymbol{x}_{k+1} + \bar{\gamma}_{k+1}],$$

and $\boldsymbol{x}_k$, $\boldsymbol{u}_k$ are the optimal trajectory and control. Hence for any other $Q \in \mathbb{S}_+^n, \{P_k \in \mathbb{S}_+^n\}_{k=1}^{\nu}$, $\eta_{1:\nu}$, $\gamma_{1:\nu}$, given a realization of the time-horizon length N, we define the "violation" of the Bellman iteration at each time instant k as

$$\psi_{k,N}(P_{k+1}, \eta_{k+1}, \xi_k, Q) = \mathop{\mathbb{E}}_{\boldsymbol{w}_k} \Big[\frac{1}{2} \boldsymbol{x}_{k+1}^T P_{k+1} \boldsymbol{x}_{k+1} + \eta_{k+1}^T \boldsymbol{x}_{k+1}$$
$$- \frac{1}{2} \boldsymbol{x}_k^T P_k \boldsymbol{x}_k - \eta_k^T \boldsymbol{x}_k + \underbrace{\frac{1}{2} \gamma_{k+1} - \frac{1}{2} \gamma_k}_{\xi_k} + \frac{1}{2} \begin{bmatrix} \boldsymbol{u}_k^T & \boldsymbol{x}_k^T & r_k^T \end{bmatrix} \mathcal{Q} \begin{bmatrix} \boldsymbol{u}_k \\ \boldsymbol{x}_k \\ r_k \end{bmatrix}\Big],$$

where $\mathcal{Q}$ are defined similarly as in (5). By Assumption II.1, it holds that $\mathbb{E}_{\boldsymbol{w}_k}[\boldsymbol{w}_k G^T P_{k+1} G \boldsymbol{w}_k] = G^T P_{k+1} G \, \mathbb{E}_{\boldsymbol{w}_k}[\boldsymbol{w}_k^2] = G^T P_{k+1} G \sigma^2$. In addition, in view of (3) and by the fact that $\boldsymbol{w}_k$ is independent of $\boldsymbol{x}_k$ and $\boldsymbol{u}_k$, we can further rewrite the "violation" $\psi_{k,N}(\cdot)$ as

$$\psi_{k,N}(\cdot) = \frac{1}{2} \begin{bmatrix} \boldsymbol{u}_k^T & \boldsymbol{x}_k^T & 1 \end{bmatrix} \underbrace{\begin{bmatrix} \mathfrak{R}_k & \mathfrak{S}_k & g_k \\ \mathfrak{S}_k^T & \mathfrak{F}_k & \beta_k \\ g_k^T & \beta_k^T & \xi_k + r_k^T Q r_k \end{bmatrix}}_{H_k} \begin{bmatrix} \boldsymbol{u}_k \\ \boldsymbol{x}_k \\ 1 \end{bmatrix},$$

where $\mathfrak{R}_k = B^T P_{k+1} B + \sigma^2 G^T P_{k+1} G + I$, $\mathfrak{S}_k = B^T P_{k+1} A$, $\mathfrak{F}_k = A^T P_{k+1} A - P_k + Q$, $g_k = B^T \eta_{k+1}$, $\beta_k = A^T \eta_{k+1} -$

$\eta_k - Qr_k$. Note that all the cross-terms regarding $\boldsymbol{w}_k$ have vanished due to Assumption II.1.

Moreover, using the Schur complement property, we can further write $\psi_{k,N}(\cdot)$ as

$$\psi_{k,N}(\cdot)=\frac{1}{2}\begin{bmatrix}\boldsymbol{u}_k^{1,T} & \boldsymbol{x}_k^T & 1\end{bmatrix}\mathfrak{G}\begin{bmatrix}\mathfrak{R}_k & \\ & H_k\backslash\mathfrak{R}_k\end{bmatrix}\mathfrak{G}^T\begin{bmatrix}\boldsymbol{u}_k^1\\ \boldsymbol{x}_k\\ 1\end{bmatrix}$$
$$=\frac{1}{2}\left\|\mathfrak{R}_k^{1/2}(\boldsymbol{u}_k^1+\mathfrak{R}_k^{-1}\begin{bmatrix}\mathfrak{S}_k & g_k\end{bmatrix}\tilde{\boldsymbol{x}}_k\right\|^2+\operatorname{tr}\left((H_k\backslash\mathfrak{R}_k)\tilde{\boldsymbol{x}}_k\tilde{\boldsymbol{x}}_k^T\right),$$

where $\mathfrak{G}_k=\begin{bmatrix}I & & \\ \mathfrak{S}_k^T\mathfrak{R}_k^{-1} & I & \\ g_k^T\mathfrak{R}_k^{-1} & & 1\end{bmatrix}$, $\tilde{\boldsymbol{x}}_k=\begin{bmatrix}\boldsymbol{x}_k\\ 1\end{bmatrix}$.

Now that if we take the expectation with respect to $\boldsymbol{x}_k|\boldsymbol{N}=N$ on $\psi_{k,N}(\cdot)$, we have

$$\mathop{\mathbb{E}}_{\boldsymbol{x}_k|\boldsymbol{N}=N}[\psi_{k,N}(\cdot)]=\operatorname{tr}\left((H_k\backslash\mathfrak{R}_k)\mathop{\mathbb{E}}_{\boldsymbol{x}_k|\boldsymbol{N}=N}[\tilde{\boldsymbol{x}}_k\tilde{\boldsymbol{x}}_k^T]\right)$$
$$+\mathop{\mathbb{E}}_{\boldsymbol{x}_k|\boldsymbol{N}=N}[\frac{1}{2}\left\|\mathfrak{R}_k^{1/2}(\boldsymbol{u}_k^1+\mathfrak{R}_k^{-1}\begin{bmatrix}\mathfrak{S}_k & g_k\end{bmatrix}\tilde{\boldsymbol{x}}_k\right\|^2].$$

Next, we introduce the domian

$$\begin{aligned}\mathscr{D}=&\{Q\in\mathbb{S}_+^n,\{P_k\in\mathbb{S}_+^n\},\eta_{1:\nu},\xi_{1:\nu-1}|P_\nu=Q,\eta_\nu=q_\nu,\\ &q_k=-Qr_k,t=1:\nu,H_k\succeq 0,k=1:\nu-1\},\end{aligned}\tag{8}$$

Notably, it holds that $H_k\backslash\mathfrak{R}_k\succeq 0$ on domain $\mathscr{D}$ since $I\succ 0$ and $\mathfrak{R}_k\succ 0$. Hence $\mathbb{E}_{\boldsymbol{x}_k|\boldsymbol{N}=N}[\psi_{k,N}(\cdot)]$ is bounded from below by zero. Now that we collect the data with different time-horizon length, we further add the above non-negative "violation" from $k=\nu-N+1$ (the time instant that a patient starts tracking) to $k=\nu-1$ and take the expectation on the random time-horizon length $\boldsymbol{N}$. This would give the expected "overall violation" of Bellman iterations at all time steps

$$\Psi(Q,P_{1:\nu},\eta_{1:\nu},\xi_{1:\nu-1})=\sum_{N=2}^{\nu}\mathbb{P}(\boldsymbol{N}=N)\sum_{k=\nu-N+1}^{\nu-1}\mathop{\mathbb{E}}_{\boldsymbol{x}_k|\boldsymbol{N}=N}[\psi_{k,N}(\cdot)].$$

On the other hand, under Assumption II.3, it is clear that $\mathbb{E}_{\boldsymbol{x}_k|\boldsymbol{N}=\nu}[\tilde{\boldsymbol{x}}_k\tilde{\boldsymbol{x}}_k^T]\succ 0$, $k=2:\nu$. In addition, by Assumption II.2, $\mathbb{P}(\boldsymbol{N}=\nu)>0$. Also recall that, for the "true" parameter $\bar{Q}$ and its corresponding $\bar{P}_{1:\nu}$,$\bar{\eta}_{1:\nu}$, $\bar{\gamma}_{1:\nu}$, the expected "overall violation" is zero. Hence if we minimize $\Psi(\cdot)$ on the domain $\mathscr{D}$ and let it attain its lower bound, we would force $\mathbb{E}_{\boldsymbol{x}_k|\boldsymbol{N}=\nu}[\psi_{k,N}(\cdot)]=0$ for all $k=1:\nu-1$. Thus making the optimal control $\boldsymbol{u}_k$ take the form of (6) and $H_k\backslash\mathfrak{R}_k=0$, $k=1:\nu-1$. In view of the expression of $H_k\backslash\mathfrak{R}_k=0$, the corresponding optimal solution $(Q^\star,P^\star_{1:\nu},\eta^\star_{1:\nu})$ would satisfy the Riccati iteration (7).

However, in practice, we can not obtain the patient's intended control $\boldsymbol{u}_k$ directly. Instead, only $\boldsymbol{\mu}_k(\omega):=\boldsymbol{u}_k(\omega)+\boldsymbol{u}_k(\omega)\boldsymbol{w}_k(\omega)$, $\omega\in\Omega$ can be measured by the one dimensional force sensor. Note once again, by Assumption II.1, $\boldsymbol{w}_k$ is zero-mean and independent of other elements, hence it holds that

$$\mathop{\mathbb{E}}_{\boldsymbol{w}_k}[(\boldsymbol{\mu}_k)^2]=\mathop{\mathbb{E}}_{\boldsymbol{w}_k}[(1+\boldsymbol{w}_k)^2](\boldsymbol{u}_k)^2=(1+\sigma^2)(\boldsymbol{u}_k)^2$$

and we can re-write $\psi_{k,N}(\cdot)$ as an expression that involves $\boldsymbol{\mu}_k$:

$$\psi^{\mu}_{k,N}(\cdot)=\mathop{\mathbb{E}}_{\boldsymbol{w}_k}\Big[\frac{1}{2}\boldsymbol{x}_{k+1}^TP_{k+1}\boldsymbol{x}_{k+1}+\eta_{k+1}^T\boldsymbol{x}_{k+1}-\frac{1}{2}\boldsymbol{x}_k^TP_k\boldsymbol{x}_k$$
$$-\eta_k^T\boldsymbol{x}_k+\xi_k+\frac{1}{2}\begin{bmatrix}\boldsymbol{\mu}_k^T & \boldsymbol{x}_k^T & r_k^T\end{bmatrix}\mathcal{Q}'\begin{bmatrix}\boldsymbol{\mu}_k\\ \boldsymbol{x}_k\\ r_k\end{bmatrix}\Big],$$

where $\mathcal{Q}'=\begin{bmatrix}\frac{I}{1+\sigma^2} & 0 & 0\\ 0 & Q & -Q\\ 0 & -Q & Q\end{bmatrix}$.

And the corresponding modified expected "overall violation" $\Psi^{\mu}(\cdot)$ follows similarly, which we will not write out explicitly due to the space limitation. As a brief summary, the constructed IOC algorithm takes the form

$$\min_{(Q,P_{1:\nu},\eta_{1:\nu},\xi_{1:\nu-1})\in\mathscr{D}}\Psi^{\boldsymbol{\mu}}(\cdot).\tag{9}$$

Nevertheless, since we do not know the actual distribution of the stochastic time-horizon length $\boldsymbol{N}$, we can not directly calculate the expectation with respect to $\boldsymbol{x}_k$ for all $k=1:\nu$. Hence the empirical mean is used to approximate the expected value. Namely, the IOC algorithm that is actually applied in practice is as follows

$$\min_{(Q,P_{1:\nu},\eta_{1:\nu},\xi_{1:\nu-1})\in\mathscr{D}(\varphi)}\Psi^{\boldsymbol{\mu}}_{\approx}(\cdot),\tag{10}$$

where

$$\Psi^{\boldsymbol{\mu}}_{\approx}(\cdot)=\frac{1}{M}\sum_{N=2}^{\nu}\sum_{i_N=1}^{M_N}\Big[\frac{1}{2}\boldsymbol{x}_\nu^{i_N,T}P_\nu\boldsymbol{x}_\nu^{i_N}+\eta_\nu^T\boldsymbol{x}_\nu^{i_N}-\frac{1}{2}\boldsymbol{x}_{\nu-N+1}^{i_N,T}$$
$$\times P_{\nu-N+1}\boldsymbol{x}_{\nu-N+1}^{i_N}-\eta_{\nu-N+1}^T\boldsymbol{x}_{\nu-N+1}^{i_N}$$
$$+\sum_{k=\nu-N+1}^{\nu-1}\left(\frac{1}{2}\begin{bmatrix}\boldsymbol{\mu}_k^{i_N,T} & \boldsymbol{x}_k^{i_N,T} & r_k^T\end{bmatrix}\mathcal{Q}'\begin{bmatrix}\boldsymbol{\mu}_k^{i_N}\\ \boldsymbol{x}_k^{i_N}\\ r_k\end{bmatrix}+\frac{1}{2}\xi_k\right)\Big],$$
$$\mathscr{D}(\varphi)=\{(Q,P_{1:\nu},\eta_{1:\nu},\xi_{1:\nu-1})\in\mathscr{D}\mid\|Q\|_F\le\varphi,\|P_k\|_F\le\varphi,$$
$$\|\eta_k\|\le\varphi,\ k=1:\nu,\ \xi_i\le\varphi,\ i=1:\nu-1\},$$

and M_N denotes the number of observations which has a time-horizon length N. It is clear that $\sum_{N=2}^{\nu}M_N=M$, where M is the total number of trials by the patient. Now we are ready to present the main theorem of this work.

Theorem III.2 (Statistical consistency) *Under Assumption II.1, II.4, II.3, II.2, let* $(Q^M,P^M_{1:\nu},\eta^M_{1:\nu},\xi^M_{1:\nu-1})$ *be the optimal solution to* (10) *constructed by* M *observed trials. It holds that* $Q^M\xrightarrow{p}\bar{Q}$ *as* $M\to\infty$.

Proof *We only sketch the proof here due to the space limitation. Part of the main idea has been introduced already. What remains to be done is to use the identifiability result in Lemma III.1 and prove the uniqueness of the optimal solution to* (9)*. Next, verify that the "uniform law of large numbers" holds and apply [23, Thm. 5.7] to prove the statement. The proof idea is similar to [12].*

IV. Simulation and Experiments

In this section, we illustrate the performance of the proposed IOC algorithm. To this end, we first identify the motor dynamics and design the admittance controller so that the patient can crank the robot by hand. Next, we illustrate the statistical consistency of the IOC algorithm by simulation (with the same physical parameters). Then we collect the patient's tracking data and use the IOC algorithm to identify her tracking motion pattern. The estimation results are also verified by real experiments.

A. Data collection phase

In the data collection phase, we ask the patient to rotate the crank to control a pointer on the screen. The goal is to let the pointer track another pointer driven by a reference signal with constant velocity, i.e., $\{r_k\}_{k=1}^{\nu}$. As mentioned in Section II, it is not physically possible for the patient to directly rotate the crank when the motor is on due to high gear ratio, we hence apply the admittance control (2) to offer the patient a smooth tactile feedback during rehabilitation training. In addition, this also provides a safety guarantee to the user.

To this end, we need to identify the coefficients m, k, b in the original system $(\hat{A}, \hat{B})$ first. We let the motor generate square-shaped torque signals, whose duty-ratio and period are uniformly randomly chosen between 50% - 80% and 2s - 5s, respectively. In addition, we also let the motor generate sinusoid torque signals, whose amplitude and frequency are $2\ N \cdot m$ and uniformly randomly selected between 0.2 - 0.5 Hz. The motor is actuated without any human interference, namely, $\boldsymbol{u}(t) = 0$ for all t and the corresponding angular positions and velocities are collected for every 0.01s. We use the `pem` function in MATLAB system identification toolbox to identify the coefficients. And consequently, the admittance controller (2) is applied using the estimated coefficients.

Furthermore, since there are inevitable identification errors in $(\hat{A}, \hat{B})$, the admittance controller (2) can never exactly lead to the desired $(\bar{A}, \bar{B})$. To mitigate this error for later IOC, we collect another set of data to identify the "actual" $(\bar{A}, \bar{B})$ after the admittance controller (2) is applied. In particular, we let the human to manipulate the crank randomly and the force sensor collects the torque input-signal $\boldsymbol{\mu}_t := \boldsymbol{u}_t + \boldsymbol{u}_t \boldsymbol{w}_t$ injected by the patient. The angular positions and velocities are collected for every 0.01s. The same identification method is used, and we have the estimate as follows

$$\bar{A} \approx \begin{bmatrix} 0 & 1 \\ 0 & -0.179 \end{bmatrix}, \quad \bar{B} \approx \begin{bmatrix} 0 \\ 55.37 \end{bmatrix}.$$

Consequently, we discretize the dynamics with a sampling peirod $\Delta t = 0.01$ and get the dynamics (A, B, G). This set of parameter would be used for later IOC.

On the other hand, we still need to identify the covariance of the process noise $\boldsymbol{w}_k$ caused by muscle-actuation. To this end, we ask the patient to rotate at a constant velocity and only those data that is closed to a "steady state" is preserved for analysis. In this case, the damping torque is also constant and the patient's "intended" torque input would be exactly the opposite of the constant damping torque $-u_d$. The constant "intended" torque input $\boldsymbol{u} = u_d$ is estimated by the mean of the force sensor measurements $\frac{1}{T}\sum_{k=1}^{T} \boldsymbol{\mu}_k(\omega)$. This is because $\boldsymbol{\mu}_k = \boldsymbol{u}_k + \boldsymbol{u}_k \boldsymbol{w}_k$ is I.I.D. over time steps in this case. By the law of large numbers, $\frac{1}{T}\sum_{k=1}^{T} \boldsymbol{\mu}_k$ would converge to the expected value $\mathbb{E}[\boldsymbol{u}_k]$. Since $\boldsymbol{u}_k = u_d$ remains constant at all time, $\mathbb{E}[\boldsymbol{u}_k] = u_d$ is exactly the constant "intended" torque input. Consequently, the variance σ^2 of $\boldsymbol{w}_k$ is estimated by sample variance of $(\boldsymbol{\mu}_k - u_d)/u_d$.

To verify if the signal-dependent noise in (3) is truly linear with respect to the control signal $\boldsymbol{u}_k$, we repeat the above experiment using different viscosity coefficients, i.e., different u_ds. Note that

$$\begin{aligned} \operatorname{cov}(\boldsymbol{\mu}_k, \boldsymbol{\mu}_k) &= \mathbb{E}[(\boldsymbol{u}_k + \boldsymbol{u}_k \boldsymbol{w}_k)^2] - (u_d)^2 \\ &= (u_d)^2 + (u_d)^2\sigma^2 - (u_d)^2 = (u_d)^2\sigma^2 \end{aligned}$$

And as can be seen from Fig. 2, the covariance estimate of $\boldsymbol{\mu}_k$ and the "intended" torque input admit a linear relation. Hence linear regression is used to estimate the variance σ^2 (the slope) and the estimate is $\sigma^2 \approx 0.00629$.

Equipped with the above set-ups, we now can start the data collection process. In particular, we let the reference signal be $r_k = [0.5k, 0.5]^T$, $k = 1 : 300$ ($\nu = 300$), namely, it rotates at a constant velocity of 50 deg/s and last for 3 seconds. We collect $M = 300$ trials of the patient.

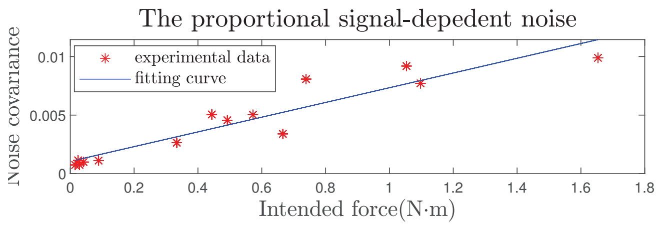

Fig. 2. The covariance o signal-depedent noise is growing with the intended force proportionally

B. Numerical simulation

To show the statistical consistency and the convergence performance of our proposed IOC algorithm, we generate 100 batches with $M = 5000$ trajectories in each batch and set $\bar{Q} = \operatorname{diag}([10, 30])$. The time instant that the patient starts tracking is randomly generated by sampling from a uniform distribution supported on $[1, 41]$. The starting angular position $\boldsymbol{x}_{k,1}^i$ is randomly chosen by sampling from a uniform distrubution supported on $[-60, 60]$ (deg), while the starting angular velocity $\boldsymbol{x}_{k,1}^i$ is set to zero for all $i = 1 : M$. For each M, we can calculate 100 relative estimate errors $\|Q^{\ell,M} - \bar{Q}\|_F / \|\bar{Q}\|_F$, one for each batch number ℓ. Since the trajectories are generated in an I.I.D. way, we can calculate the empirical mean and standard deviation of the relative estimate error. As can be seen from Fig. 3, in a log-log plot, the logarithm of the relative error mean and the standard deviation decreases as $\log(M)$ increases and are also approximately affine to $\log(M)$.

We further run an affine fit two the logarithm data and get that `Mean of relative error` $\approx \mathcal{O}(M^{-0.4119})$, `Standard deviation of relative error` $\approx \mathcal{O}(M^{-0.4635})$. We hence suspect that the convergence rate is $\mathcal{O}(M^{-0.5})$, like most M-estimators such as maximum log-likelihood [23, p. 51]. Theoretical analysis regarding this convergence rate would be left for future work.

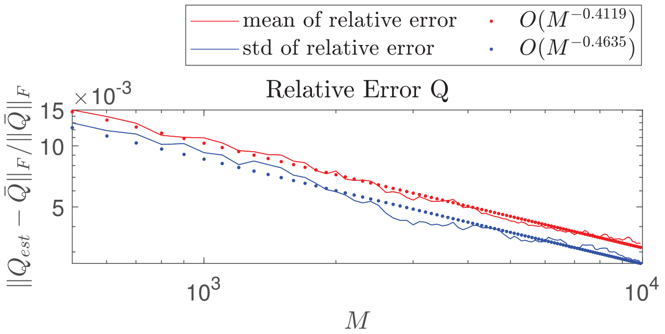

Fig. 3. Log-log plot of the mean and standard deviation of the relative error of the estimate as a function of the number of trajectories.

C. Verification for the IOC estimate

The performance of the IOC is verified by experiments. In particular, we let the starting angular position to be the same ($\boldsymbol{x}_0 = [0, 0]^T$) for all trials and collect 100 trajectories. We further separate them into groups depending on their time-horizon length and choose the group that contains the most trajectories, i.e., the one with the largest M_N. Next, we verify the identified model by Monte-Carlo simulations. In particular, the "forward" problem model (5) are simulated based on the estimated $\bar{Q}$. The comparison between the Monte-Carlo simulation results and the actual measured trajectories ($M_N = 5$) with the same time-horizon length ($N = 264$) are illustrated in Fig. 4. As can be seen in the figure, the identified model predicts the motion of the crank under the patient's manipulation well.

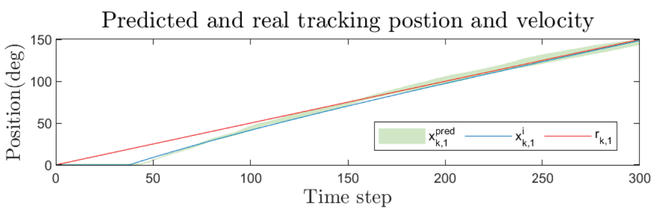

Fig. 4. Illustration of predicted state trajectories v.s. the actual measured ones.

For further work, the personalized motion model could be used in training or diagnosis for hemiplegia recovery.

V. Conclusion

In this work, we have developed the inverse optimal control algorithm to identify the personalized tracking motion pattern for hemiplegia recovery. We prove that the model structure is strictly globally identifiable and the statistically consistent IOC algorithm can deal with signal-dependent noise and stochastic time-horizon length. Moreover, the crank's system dynamics and the covariance of multiplicative noise induced by patients' muscle fluctuations are estimated. Based on the identified parameters, the performance of the IOC algorithm is tested and verified by simulations and experiments.

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